

Assessing Distraction Potential of Augmented Reality Head-Up Displays for Vehicle Drivers

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Objective: To develop a framework for quantifying the visual and cognitive distraction potential of augmented reality (AR) head-up displays (HUDs).

Background: AR HUDs promise to be less distracting than traditional in-vehicle displays because they project information onto the driver's forward-looking view of the road. However, AR graphics may direct the driver's attention away from critical road elements. Moreover, current in-vehicle device assessment methods, which are based on eyes-off-road time measures, cannot capture this unique challenge.

Method: This article proposes a new method for the assessment of AR HUDs by measuring driver gaze behavior, situation awareness, confidence, and workload. An experimental user study ($n = 24$) was conducted in a driving simulator to apply the proposed method for the assessment of two AR pedestrian collision warning (PCW) design alternatives.

Results: Only one of the two tested AR interfaces improved driver awareness of pedestrians without visually and cognitively distracting drivers from other road elements that were not augmented by the display but still critical for safe driving.

Conclusion: Our initial human-subject experiment demonstrated the potential of the proposed method in quantifying both positive and negative consequences of AR HUDs on driver cognitive processes. More importantly, the study suggests that AR interfaces can be informative or distractive depending on the perceptual forms of graphical elements presented on the displays.

Application: The proposed methods can be applied by designers of in-vehicle AR HUD interfaces and be leveraged by designers of AR user interfaces in general.

Keywords: augmented reality, interface evaluation, situation awareness, driver distraction

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INTRODUCTION

Distracted driving has significant consequences on transportation safety. On U.S. roadways in 2015, distraction-affected crashes accounted for 14% of all police-reported crashes and 10% of fatal crashes, resulting in 3,477 deaths (National Center for Statistics and Analysis, 2017). This phenomenon may be under-reported because postcrash investigations often rely on witness accounts or drivers' self-reporting their negative behavior. In fact, a recent analysis of a large-scale naturalistic driving study data set (Dingus et al., 2016), which captured detailed driving contexts during 905 crashes, revealed that driver distraction accounted for 68.3% of police-reportable crashes. Surprisingly, more than 50% of the time while driving, people were engaged in distracting activities such as interacting with passengers, handheld cell phones, and in-vehicle devices.

As carried-in devices (e.g., cell phones) and in-vehicle information systems (IVIS) are getting more popular and complicated, they become continued sources of driver distraction. A recent experimental study investigated driver distraction associated with IVIS in 30 vehicles (Strayer et al., 2017). They examined the demand imposed by IVIS interactions compared to high-demand referent tasks such as a concurrent visual surrogate reference task and an auditory *N*-back task. In about 77% of vehicles, IVIS interactions imposed equal or even higher demand than the high-demand referent tasks. The findings also showed that interacting with in-vehicle devices often places demands on multiple resources and distracts drivers visually, manually, and/or cognitively from the primary task of driving.

Augmented reality (AR) is one of the promising solutions to address the distraction potential of current IVIS. By projecting computer-generated graphics onto the driver's forward

field of view, AR head-up displays (HUDs) allow drivers to perceive information without shifting attention away from the road. In a driving simulator study, Medenica, Kun, Paek, and Palinko (2011) showed benefits of an AR navigation aid over a traditional center-stack display. With the AR HUD, drivers kept their gaze on the road longer, reacted faster to hazards, and improved driving performance, with lower mental workload. AR HUDs can be even further beneficial to drivers when the information is directly overlaid on the real-world object it refers to (known as conformal presentation). Kim, Gabbard, Anon, and Misu (2018) demonstrated the benefits of conformal presentation over the traditional screen-fixed presentation, which displays symbols at a fixed location on the HUD. Conformal graphics guided drivers' attention quickly and had positive consequences on driver behavior and performance.

However, empirical studies have also reported possible negative consequences of HUDs, in part due to AR graphics' salience (Sharfi & Shinar, 2014), frequent changes (Wolffsohn, 1998), and visual clutter (Burnett & Donkor, 2012). For example, Sharfi and Shinar reported that highlighting lane markers in foggy or night driving conditions had positive effects on drivers' confidence and workload but impaired their ability to detect pedestrians. Wolffsohn examined driver attention by asking drivers to read a HUD speedometer aloud while attending to the brake lights of a lead vehicle and found that frequently changing information on the HUD captured drivers' attention and caused inattention blindness even within their central vision. These findings imply that AR HUDs may distract drivers from critical road elements without taking their eyes off the road or engaging them in any tasks irrelevant to driving.

Even worse, current in-vehicle device assessment methods may not capture the aforementioned challenges found in AR HUDs. The National Highway Traffic Safety Administration (NHTSA; 2012) has published driver distraction guidelines, including a set of procedures, to assess the *visual and manual* distraction potential of in-vehicle devices. However, the proposed acceptance criteria for IVIS interaction are based on eyes-off-road-time measures and cannot capture the inattention blindness caused by AR HUDs.

To assess *cognitive* distraction in the automobile, Strayer et al. (2015) proposed a framework to standardize the cognitive workload imposed by typical secondary tasks, including listening to the radio and audiobooks, as well as conversations with passengers and on a hands-free phone. Recently, the framework was extended to quantify the multimodal demands of IVIS interactions for secondary tasks such as making a phone call, sending a text, tuning the radio, and programming navigation systems (Strayer et al., 2017). They evaluated drivers' deteriorated ability to detect visual or tactile stimuli while using IVIS. Note that the assessment method was operationalized such that participants were overtly asked to conduct concurrent tasks while driving because the framework focuses on quantifying workload imposed by secondary tasks. This method may not be applicable to AR HUDs that support the primary task of driving (e.g., highlighting lane markers) because drivers are not engaged in any secondary tasks.

Objectives and Approaches

Given the possible negative consequences of AR HUDs and limitations of current assessment methods, we need to develop up-to-date methods specifically valid for AR HUDs with the goal of informing drivers while minimizing distraction. Thus, this work aimed to (1) develop a new method for quantifying the visual and cognitive distraction potential of AR HUDs and (2) demonstrate the potential of the proposed method in a human-subject experiment. Specifically, this article proposes a method that evaluates the effects of AR HUDs on driver cognitive processes by measuring driver visual attention, situation awareness (SA), confidence, and workload. To quantify the visual distraction potential, we compared driver gaze behavior with and without using AR HUDs. Instead of using traditional eyes-off-road-time measures, we analyzed driver gaze distribution across different road elements. To quantify the cognitive distraction potential, we went beyond the previous method (Strayer et al., 2017), which compares drivers' ability to *detect* artificial stimuli with and without using IVIS. Appreciating the spatiotemporal demands of driving, we instead measured different levels of drivers' ability to not only *detect* but also *understand* and *predict* the dynamic driving

environment. Our method is based on Endsley's definition of SA, which includes three levels of awareness such as "the perception of the elements in the environment (L1), the comprehension of their meaning (L2), and the projection of their status in the near future (L3)" (Endsley & Jones, 2011). Finally, to demonstrate the potential benefits of the proposed method, we conducted a human-subject experiment in a driving simulator and applied the method to evaluate two candidate AR HUD interfaces.

Selected Use Case and AR HUD Interfaces

On average, a pedestrian was killed every 1.5 hr on U.S. roadways in 2016 (National Center for Statistics and Analysis, 2018). Over the last decade in the United States, pedestrian deaths have increased (e.g., 4,699 in 2007 and 5,987 in 2016) while the total number of traffic fatalities has decreased (e.g., 41,259 in 2007 and 37,461 in 2016). Pedestrian deaths accounted for 16% of the U.S. traffic fatalities and the rates are significantly higher in major cities (e.g., 59.6% of traffic fatalities were pedestrians in New York City) in 2016. Even worse, about 25.8% of pedestrians were killed or incapacitated, which is significantly higher than the 7.1% rate of vehicle occupants in 2015 on U.S. roadways (NHTSA, 2017). Moreover, current in-vehicle pedestrian collision warnings (PCWs) often lack spatial information, which is critical for the driver's appropriate reaction.

To address the limitations of current warnings, our prior work (Kim, Isleib, & Gabbard, 2016) proposed a novel AR PCW, the *virtual shadow*. Usability experts evaluated the virtual shadow as well as an extant PCW, the *bounding box* (Figure 1), and predicted that:

- Virtual shadows would not narrow, but guide driver attention to pedestrians; help drivers identify dangerous pedestrians and predict their movement; and reduce the workload.
- Bounding boxes would guide but also narrow driver attention to pedestrians; help drivers detect pedestrians but impair the ability to identify and predict dangerous pedestrian movements; and increase the workload.

In this study, we tested driver behavior and performance with AR PCWs in a driving simulator to see if the benefits predicted by usability experts are observed in an empirical user study. For more information about the virtual shadow design and expert evaluation process, refer to Kim et al. (2016).

METHOD

Participants

Adapting NHTSA's driver distraction guidelines (NHTSA, 2012), we recruited a gender-balanced sample of 24 licensed drivers with normal or corrected-to-normal vision. On average, participants were 21.4 years old ($SD = 3.1$) with 4.7 years of driving experience ($SD = 2.9$).

Apparatus

A medium-fidelity driving simulator with augmented driving video footage was used to provide our participants with an immersive and naturalistic driving experience (Figure 2). We captured driving video footage from real-world driving. Each video was about 10 min long and contained various road events, including pedestrian crossings, braking lead vehicles, and changing speed limits. From this set, we selected three driving videos that are similar in terms of road complexity and the number of road events but with different scenarios in terms of specific routes and road events. For rapid prototyping of AR interfaces, we then used a video-editing tool to overlay computer-generated graphics (i.e., the AR graphics) atop the driving videos. The augmented driving videos were then projected onto a large curved wall in front of the half-cab ($\sim 130^\circ$ field of view) along with a small crosshair that was controlled by the steering wheel. We also provided a speedometer synchronized with the driving footage. SMI eye-tracking glasses recorded participants' gaze behavior at 30 Hz.

Procedure and Tasks for Participants

The user study consisted of four parts: (1) pretest survey, (2) a practice trial, (3) experimental trials, and, (4) a posttest interview. After a brief introduction and informed

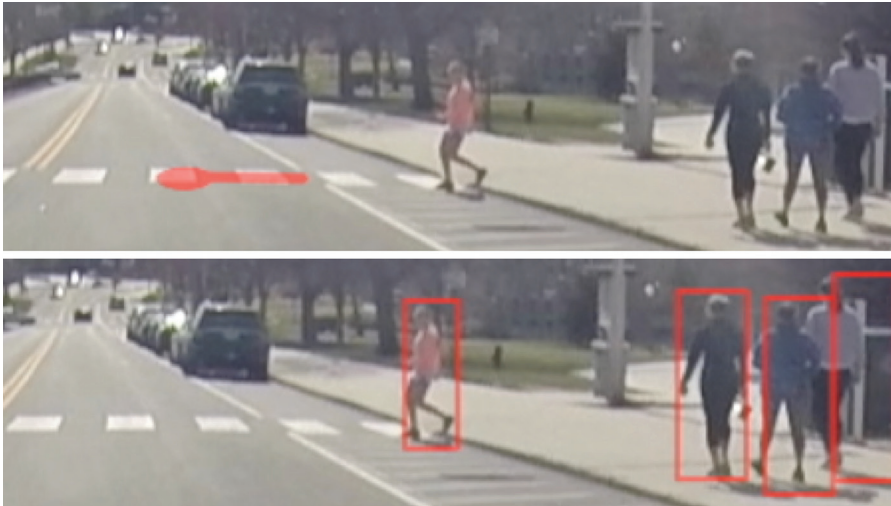


Figure 1. Augmented reality (AR) pedestrian collision warnings selected for testing. The virtual shadow (top) shows a ground-level projection of a pedestrian's anticipated path. The location of the circle represents the predicted location of collision, the direction of line indicates the direction from which a pedestrian is approaching, and the length of the line represents the spatial intrusion by the pedestrian which is changing over time depending on its relative movement. Bounding boxes (bottom) highlight all detected pedestrians (Kim et al., 2016).

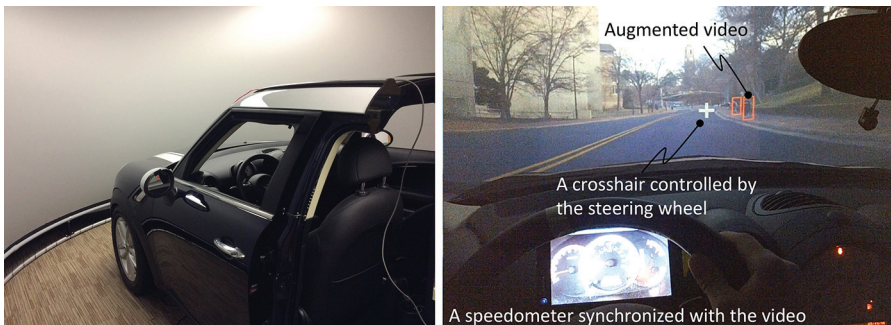


Figure 2. A driving simulator with a real-car cab and a large wall-projection screen (left) was used to present augmented driving videos (right, captured from the driver's perspective).

consent, we surveyed participants' demographic information and driving experience. In a practice trial, an experimenter gave an in-car orientation to explain simulator details. One practice drive was given to allow participants time to get familiar with the driving simulator and the task procedure. In experimental trials, each participant had three drives. Participants were asked to perform wayfinding tasks while

experiencing different AR PCWs. During each drive, the simulator was paused at random time points, and the experimenter asked questions about the situation just experienced as well as the participant's confidence in each answer. After each drive, participants gave subjective ratings on their workload. In posttest interviews, participants gave comments on the AR PCWs.

Participant	Trip1	Trip2	Trip3
P1	A γ	B β	C α
P2	B α	C γ	A β
P3	C β	A α	B γ

Figure 3. An example of a 3 $\times$ 3 Graeco-Latin square. Each participant experienced three counter-balanced ordered pairs of warnings {A, B, C} and driving scenarios { α , β , γ }, such that each participant saw each warning interface and driving scenario exactly once.

Participants' tasks were operationalized in the video-based simulation environment as follows. First, we framed the participants' primary task as *wayfinding* in unfamiliar areas to simulate the cognitive demand of driving. For each drive, participants were asked to navigate to a specific destination using a map-based route plan and turn-by-turn voice instructions via pre-recorded audio. As part of our driver SA measurement, we asked questions about landmarks (e.g., street names) that participants would need to know for navigational success. To approximate the visual and manual demands of driving, we required participants to *mimic the driving* conveyed in the videos. Using the steering wheel as a control, participants were instructed to match the crosshair at the center of the driving lane. No perturbations were introduced for the crosshair matching task. We also asked participants to manipulate pedals to mimic the longitudinal movement and speed of the vehicle prescribed in the video and the speedometer. Even though pedal operations did not directly control video presentation, participants were told that driving performance was measured by how well they matched the vehicle movement prescribed in the video. During the practice drive, an experimenter monitored participants' pedal and steering operations via simulator logging software and gave corrective feedback when needed. We did not give any specific instruction for participants regarding the pedestrian warnings.

Experimental Design

A repeated-measures experiment was conducted such that each participant had three

trips to experience different warning conditions. In addition to two AR PCW conditions (i.e., virtual shadow and bounding box), we added a no warning condition as a baseline for driver behavior and performance measures. The presentation order of both warning conditions and driving scenarios were counter-balanced by Graeco-Latin squares (Martin & Nadarajah, 2014) so that each participant experienced all three warning conditions in different driving scenarios (Figure 3).

Measures

To quantify the visual and cognitive distraction potential of AR HUDs, we established the following dependent measures related to drivers' visual attention (gaze behavior) and cognitive processes (SA, overconfidence bias, and workload).

Gaze behavior. We analyzed driver gaze behavior within a 10-s time window right before each pause to characterize the allocation of the driver's visual attention. The 10-s window was selected from the visible time of objects relevant to SA questions (maximum of 9.5 s captured in our videos) and the time drivers may need to obtain SA (7–12 s; Lu, Coster, & de Winter, 2017). Using SMI's BeGaze 3.5 software, we assigned participants' eye glances to areas of interest (AOIs), such as pedestrians, vehicles, roadways, traffic signs, and the dashboard. Driver gaze patterns were then quantified by gaze distribution (% dwell time), which measures the proportion of time participants' gaze resided on each AOI. Note that we used a dynamic AOI for each object on the road, instead of the unitary and static AOI for the windshield as is typical in traditional driver gaze analysis for eyes-off-road time measures.

SA. Driver SA was measured by the widely accepted Situation Awareness Global Assessment Technique (SAGAT; Endsley & Jones, 2011) with some operational modifications (see next section for details). SA probes were developed from a goal-directed task analysis (GDTA; Endsley & Jones, 2011), which hierarchically identified drivers' goals, decisions to be made to achieve each goal, and required information for making each decision. SA probes specific to



Figure 4. An example of the perceptual matching technique. To measure driver awareness of nearby vehicles (top), we paused the scenario, presented a picture of the empty scene (bottom), and asked participants to tell the numbers in the given picture to locate vehicles.

each driving context were then developed based on information requirements. The outcomes from GDTA and SA queries were reviewed by four subject matter experts to ensure content validity. After data collection, the accuracy of each answer to a given question was evaluated against the ground truth of the driving scenario and quantified by a binary *SA score* (correct or incorrect).

Overconfidence bias. Participants were asked to rate their confidence (high or low) in each answer to the SA probes. A binary *overconfidence score* was assigned by relating the participant's subjective confidence and objective accuracy of SA, such that high confidences in incorrect answers were considered overconfident (Sulistyawati, Wickens, & Chui, 2011). For example, we assigned an overconfidence score of 1 if a participant misperceived the current state of a traffic light with high confidence.

Workload. Driver workload was measured by the NASA Task Load Index (TLX; Hart, 2006), which is a subjective assessment tool that considers six dimensions of workload (overall performance, frustration, effort, mental, physical, and temporal demand of a task).

Qualitative feedback on AR HUD interfaces. We also collected participants' qualitative comments on their experience with each AR HUD interface via unstructured interviews.

Operational Modifications on SAGAT

The first modification was to employ a *perceptual matching technique* (Figure 4). To directly measure operator SA, SAGAT asks questions about a dynamic situation while visually blacking out the task scenario unexpectedly. The operator's answers are then objectively evaluated against ground truth. However, some researchers have argued that SAGAT may be a test of visual memory capacity in front of a blank screen (Durso & Dattel, 2004). To help mitigate this drawback and facilitate the querying procedure, at each pause, we presented a picture of an empty driving scene (same as the driving scenario without the road actors) instead of a blank screen. Participants were asked to perceptually match locations of specific road actors with numbers presented in the picture.

The second modification was to measure not only the level (or depth) but also breadth of driver knowledge about the dynamic driving environment (Figure 5). For this purpose, all SA probes were classified into four groups by different types of environmental constraints identified from a work domain analysis (Kim et al., 2016): pedestrians, moving (e.g., other vehicles), stationary (e.g., road geometry and landmarks), and legal obstacles (e.g., traffic signals and road signs). We then identified multiple freezes in a given video and presented four randomly selected

		Breadth of driver knowledge about the situation			
Depth of driver knowledge		Pedestrian (cued by AR)	Moving Obstacles (other vehicles, un-cued)	Stationary Obstacles (landmarks, un-cued)	Legal Obstacles (signals/signs, un-cued)
					
	L1 Perception	Q11. Detect location?	Q21. Detect location?	Q31. Detect landmarks?	Q41. Detect signals?
	L2 Comprehension	Q12. Judge convergence?	Q22. Judge convergence?	Q32. Current ego-position?	Q42. Current state/meaning?
	L3 Projection	Q13. Judge collision?	Q23. Judge collision?	Q33. Next navigational choice	Q43. Future state/affordance?

Figure 5. Drivers were queried about objects not only cued but also un-cued by augmented reality head-up displays (AR HUDs). For each trip, the driving scenario unexpectedly paused four times: one for each type of environmental constraint. At each pause, we asked three questions to measure the level of driver awareness of a selected environmental constraint. Consequently, for each trip, a total of 12 questions were asked to assess both *breadth* (four types of environmental constraints) and *depth* (three levels) of driver knowledge about the dynamic driving environment. See Table A1 for specific questions.

freezes such that SA probes in a drive covered all four types of environmental constraints. Even though AR warnings were always augmenting pedestrians before the pauses, participants were sometimes asked about other critical scene elements, such as other vehicles or road signs. Therefore, participants could not predict when the scenario would freeze and what environmental elements would be queried. Furthermore, at each pause, we asked three questions about a selected environmental constraint to measure the driver’s level of SA. For example, to assess the level of driver awareness of pedestrians, we asked three questions to see if the driver was able to detect nearby pedestrians (L1 perception), identify dangerous ones among others (L2 comprehension), and predict their movement relative to their own vehicle (L3 projection).

Data Analysis

A repeated-measures analysis of variance (ANOVA) was conducted on gaze dwell time. We performed Cochran’s *Q* tests on SA and overconfidence scores (binary) and Friedman tests on TLX scores (ordinal), which are non-parametric alternatives (Conover, 1999) to the repeated-measures ANOVA. Whenever we found significant main effects, we performed appropriate post hoc tests depending on the types of dependent variables, such as paired *t*-tests or McNemar tests with Tukey’s

adjustment or false discovery rate adjustments to account for multiple comparisons (Bretz, Westfall, & Hothorn, 2016).

RESULTS

We analyzed data from 23 participants, excluding one who experienced motion sickness in the driving simulator. To quantify the distraction potential of a given AR HUD, we focused on *changes* in driver behavior and performance as compared to the control (no warning) condition.

Driver Gaze Behavior

Eyes-off-road times were very low in all conditions (2.33%, 2.42%, and 3.24% of the time with virtual shadow, bounding box, and no warning, respectively) and differences between conditions were not statistically significant. However, AR warnings had significant effects on driver gaze distribution across road elements (Figure 6), especially gaze on pedestrians, $F(2, 42) = 101.39$, $p < .001$, and legal obstacles, $F(2, 42) = 4.10$, $p = .02$. Post hoc tests revealed that drivers looked at pedestrians more with bounding boxes (25.8% of the time) while spending less time looking at legal obstacles (6.35% of the time).

Driver SA

Cochran’s *Q* tests revealed significant effects of AR warnings on driver awareness of

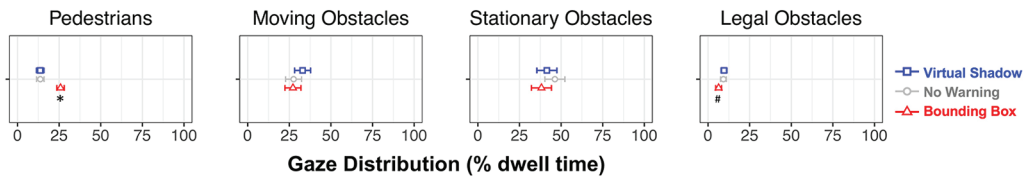


Figure 6. Allocation of driver visual attention measured by percent dwell time on four types of environmental elements. Error bars represent 95% confidence intervals. The significant changes are reported as compared to the no warning condition by $*p < .05$, $\#p < .1$ from post hoc paired t -tests.

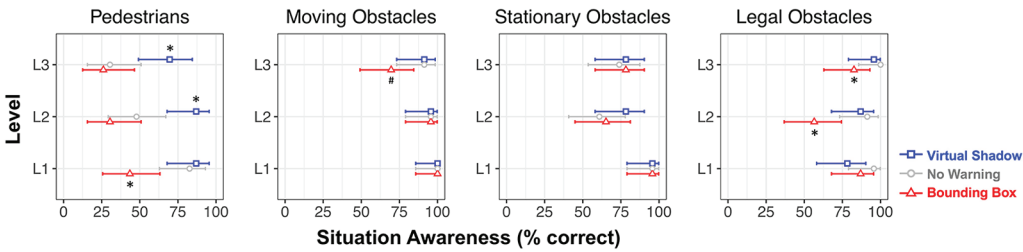


Figure 7. Mean proportion (%) of correct answers are reported as situation awareness (SA) performance indices which are decomposed into three levels (L1 perception, L2 comprehension, and L3 projection) of driver awareness of four types of environmental constraints. Error bars represent 95% modified Wilson binomial confidence intervals (DasGupta, Cai, & Brown, 2001). The significant changes in SA as compared to the no warning condition are reported by $*p < .05$, $\#p < .1$ from post hoc McNemar tests.

pedestrians (L1, L2, and L3) and legal obstacles (L2 and L3). The effect on driver awareness of moving obstacles (L3) was marginal. No significant effect was found on awareness of stationary obstacles (Figure 7).

Driver awareness of pedestrians varied across warning conditions; $Q(2, 23) = 10.706$, $p = .005$ for perception, $Q(2, 23) = 12.667$, $p = .002$ for comprehension, and $Q(2, 23) = 10.111$, $p = .006$ for projection. With virtual shadows, participants more correctly identified pedestrians who were heading into the vehicle's path (L2) and predicted possible collisions (L3), compared to the no warning condition. With bounding boxes, participants less correctly answered questions about their perception of pedestrians (L1).

Regarding moving obstacles, participants correctly predicted (L3) other vehicles' relative movement (e.g., whether the headway from a lead vehicle was decreasing or not) 91.30% of the time in both the no warning and virtual shadow conditions and 69.57% in the bounding box condition. However, the effects of visual warnings on

participants' awareness of moving obstacles were marginal, $Q(2, 23) = 5.329$, $p = .07$.

Visual warnings also had impacts on driver awareness of legal obstacles such as traffic lights and signs, $Q(2, 23) = 8.143$, $p = .017$, for comprehension, and $Q(2, 23) = 6.5$, $p = .039$ for projection. With bounding boxes, participants less correctly understood (L2) consequences of the perceived signs (e.g., whether they were speeding or not when a speed limit sign was visible) and predicted (L3) the situation changes in time (e.g., whether they can make it through the intersection or not when a traffic light was visible).

Overconfidence in SA

We found significant effects of AR warnings on overconfidence in driver awareness of pedestrians (L1, L2, and L3) and moving obstacles (L3). No significant effect was found on overconfidence in legal and stationary obstacles (Figure 8).

There were significant effects of visual warnings on overconfidence in driver awareness of

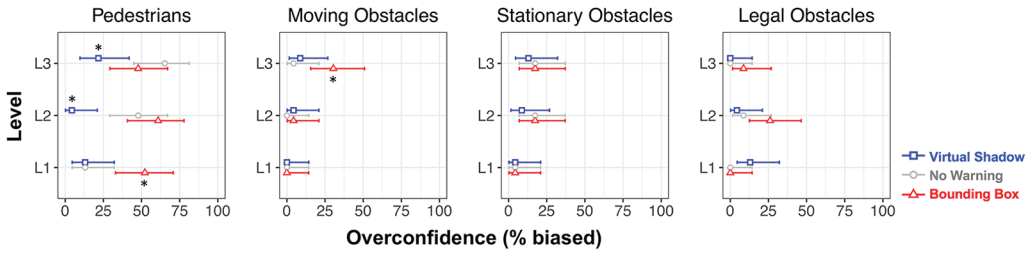


Figure 8. Mean proportion (%) of overconfident answers are reported as overconfidence scores, which are decomposed into three levels (L1 perception, L2 comprehension, and L3 projection) of driver awareness of four types of environmental constraints. Error bars represent 95% modified Wilson binomial confidence intervals. The significant changes in overconfidence as compared to the no warning condition are reported by * $p < .05$, # $p < .1$ from post hoc McNemar tests.

pedestrians, $Q(2, 23) = 10.800$, $p = .005$ for perception, $Q(2, 23) = 14.632$, $p = .001$ for comprehension, and $Q(2, 23) = 8.941$, $p = .011$ for projection. Virtual shadows significantly reduced participants' overconfidence in comprehension and projection while bounding boxes increased overconfidence in the perception of pedestrians.

AR warnings had impact on participants' overconfidence in their prediction of other vehicles' relative movement (L3), $Q(2, 23) = 6.200$, $p = .045$. Specifically with bounding boxes, participants were overconfident in their ability to predict other vehicle's movement 30.43% of the time, which was higher than 4.35% of the time without warning.

Workload

No effect of visual warnings on participants' workload was found, $\chi^2(2) = 2.782$, $p = .249$. The reported median TLX scores (out of 100) were 45.0, 46.7, and 40.0 with virtual shadow, bounding box, and no warning conditions, respectively. Friedman's tests on all six subscales of the TLX also found no effect of visual warnings.

DISCUSSION

The empirical evaluation partially supported the usability experts' predictions and provides further insight into driver cognitive processes with AR HUDs. As the main purpose of this work is to develop a new method for AR HUD evaluation, we first discuss the implications of

these findings for AR interface evaluation, followed by insights for interface design and the contributions and limitations of the study.

Implications for AR HUD Interface Evaluation

The human-subject experiment suggests that AR HUDs have an impact on driver visual attention allocation, but eyes-off-road time measures are insufficient to capture the impact. Virtual shadows did not affect drivers' overall pattern of visual attention allocation (Figure 6). Bounding boxes, on the other hand, attracted driver visual attention more (+12.14%) to pedestrians at the expense of reduced visual attention to other environmental elements such as landmarks (-8.07%), traffic signs (-2.88%), dashboard (-0.82%), and vehicles (-0.37%). However, both warnings may not be considered distractive according to traditional eyes-off-road criteria. These results suggest a need for refined criteria for "visual distraction" (Recarte & Nunes, 2008) to capture the unique visual behavior when using AR HUDs. By considering different types of objects on the road as separate and dynamic AOIs, we showed a better way to analyze driver gaze behavior for understanding and quantifying the visual distraction potential of AR HUDs.

To capture the inattention blindness problem found in AR HUDs, we need to examine not only driver gaze behavior but also SA. In our experiment, with bounding boxes, drivers looked at pedestrians more but detected pedestrians less

correctly. Participants were also less aware of other objects on the road still visible in their forward field of view. An interesting observation was that bounding boxes did not affect drivers' *detection* of other road elements, but rather degraded *higher-level SA* (Figure 7). For example, drivers were able to detect speed limit signs or the lead vehicle ahead (L1) but less likely understood the meaning of the sign (e.g., if they were speeding, L2) or predicted the lead vehicle's movement in time (e.g., if the gap from the lead was getting smaller, L3). The traditional detection reaction task (Strayer et al., 2017), which examines drivers' ability to detect stimuli while interacting with IVIS, may not capture this challenge.

More importantly, the human-subject experiment suggests that SA may be one of the promising measures for assessing not only how informative but also how distractive AR interfaces are (Figure 5). To quantify *how informative* AR warnings are, we examined the *depth* of driver knowledge about the environmental elements (i.e., pedestrians) *cued* by the HUD. To quantify *how distractive* AR warnings are, we investigated the *breadth* of driver knowledge about the environmental elements *un-cued* by the HUD (e.g., other vehicles, landmarks, and road signs). By doing so, we revealed that virtual shadows improved driver awareness of pedestrians while not affecting awareness of other road elements. Bounding boxes appeared to be not informative but distractive. Note that both warnings may not be considered cognitively distractive because they do not overtly involve drivers in any secondary tasks that take their mind off the driving (Recarte & Nunes, 2008; Regan, Lee, & Young, 2008; Strayer et al., 2015). This observation calls for refined criteria for "cognitive distraction" to capture the unique challenges found in AR HUDs. This insight resonates with some of the recent definitions which consider driver distraction as misallocation of attention (Smiley, 2005) or mismatch between driver resource allocation and task demands (Engström et al., 2013) regardless of the presence of secondary tasks competing for the limited attentional resources.

Contrary to the expert usability evaluation, the user study did not provide any evidence for reduced driver workload when AR warnings were given. This might be, in part, because the

mimic driving task itself was too easy (the TLX score was 40 out of 100 in the control condition). The finding needs to be confirmed by future studies in interactive driving simulators and on real roadways. Another possible reason for this finding is that subjective mental workload measures might not be sensitive enough to detect differences when operators perform the same task with slightly different user interfaces presented on the same display (Angell et al., 2006). Future work could include objective measures as well.

One of the new insights from the empirical evaluation is that overconfidence in SA is yet another important indicator of the usability of AR interfaces. In closed-loop interactions between human and environment, metacognition plays a critical role (Sethumadhavan, 2011; Sulistyawati et al., 2011). In the driving context, drivers' belief about their own SA will direct their subsequent actions to seek (or not seek) more evidence prior to acting. In our study, participants were overconfident in their ability to detect pedestrians with bounding boxes, suggesting that AR interfaces could affect users' overconfidence and might have significant consequences on their performance and safety.

Implications for AR HUD Interface Design

Conformal presentation of AR graphics is known to be beneficial to vehicle drivers (Merenda et al., 2018) and airplane pilots (Wickens, Hollands, Banbury, & Parasuraman, 2015). However, our results suggest that conformal presentations can be either informative or distractive depending on the perceptual forms of interface elements. This observation provides further insight into how to present AR graphics to better support driver SA and confidence.

In fact, the bounding box can be considered as a technology-driven interface design as it is a typical representation of detected pedestrians in machine-vision algorithms (Zhang, Benenson, Omran, Hosang, & Schiele, 2016), which may not consider human factors. The experiment showed that bounding boxes did not help with driver awareness of pedestrians. The posttest interview with participants revealed that highlighting all detected pedestrians was considered clutter and even prevented participants from

perceiving fine details of pedestrians, such as heading and posture, that might be critical for their prediction of pedestrians' movement. The results also showed negative side effects of bounding boxes, such as degraded driver awareness of un-cued scene elements. Combined with findings from the driver gaze analysis, this observation implies that bounding boxes may direct too much visual attention to pedestrians so that other critical objects are left unattended or less-attended (Figure 6). Consequently, the warning prevented the further processing (e.g., perception, comprehension, and projection) of other visual input due to lack of attentional resources. This observation also resonates with the well-known trade-off between benefit (better selective attention) and cost (worse divided attention) of attentional guidance (Wickens et al., 2015).

Conversely, the virtual shadow achieved these two contradicting goals by cueing only pedestrians who are expected to intrude the vehicle's path. Virtual shadows improved participants' awareness of pedestrians while not affecting their awareness of other road elements. More importantly, the novel design would allow drivers to accurately predict possible collisions by visualizing the invisible mechanism of collision (i.e., the dynamics of pedestrians' relative movement) in the form of a familiar shadow metaphor. This may support drivers to develop an accurate mental model and reduce overconfidence in their own SA. Moreover, drivers could do so by relying on a lower-level perceptual process rather than high-level analytic mental computation (Bennett & Flach, 2011). Therefore, more attentional resources may be reserved for drivers to deploy their attention broadly across other environmental elements.

Contributions, Limitations, and Future Work

In this article, we proposed a new framework for quantifying both the positive and negative consequences of AR HUDs on driver cognitive processes that may not be captured by current in-vehicle device assessment methods. Dynamic AOIs were used to examine driver gaze distribution across objects on the road to quantify the visual distraction potential of AR HUDs. We proposed a perceptual matching technique and extended SAGAT to measure both the depth and

breadth of driver knowledge about the driving environment and assess the degree to which AR HUD designs are informative and/or distractive. The human-subject experiment in a driving simulator demonstrated the potential of the proposed set of methods and provided insights into AR interface design and evaluation.

Despite the aforementioned contributions, this work also has limitations to generalize the findings to other situations. The most obvious limitation is the video-based driving simulation where participants pretend to drive while manipulating the steering wheel and pedals to match the movement of the car recorded in the video. The crosshair matching task might affect participant visual attention allocation and the open-loop control of the vehicle movement may not be translated into the closed-loop control in real-world driving. Therefore, the proposed set of methods would better be suited, especially at the early stage of AR interface development, for rapid prototyping and comparative evaluation among design alternatives.

Moreover, the proposed modification of SAGAT could be applied in simulated environments but problematic in field studies because it still requires freezing the task scenario. This challenge might be attenuated to some extent by additional operational modifications, such as querying when the car stops in traffic (Sirkin, Martelaro, Johns, & Ju, 2017). The findings from the young participants may not be observed in an older population, so future work could consider an age-balanced sample of participants as proposed in NHTSA's guideline (NHTSA, 2012).

Even though the framework was developed for AR HUDs in transportation, it could be extended to usability evaluation of AR applications that requires users' movement in dynamic environments. In particular, the method for quantifying both depth and breadth of the user's awareness of dynamic situations can be applied in most simulated AR environments and could assist designers of AR applications in determining design and safety trade-offs. Finally, our initial user study just demonstrates the potential of the proposed framework. The findings from this study and value of the proposed techniques need to be replicated or otherwise validated by large-scale future studies.

APPENDICES

TABLE A1: Examples of SA Questions

	Pedestrians	Moving Obstacles	Stationary Obstacles	Legal Obstacles
L1 Perception	Are there any pedestrians around you? If any, please select their locations from the given picture.	Do you have a car ahead of you? If you have any vehicles around you, please select their locations from the given picture.	Which lane are you in? (1) left, (2) right, (3) center	What is the current speed limit of the road? What is the color of the traffic light for you? What is your current speed?
L2 Comprehension	Among pedestrians you chose, are there any pedestrians heading into or currently on your path? If any, please select their locations from the given picture.	Is the car ahead of you going to change lane or make a turn? If you have any vehicles heading into or currently on your path, please select their locations from the given picture.	Which street are you on? Locate yourself on the given map.	Are you traveling (1) below, (2) above, (3) about the same (± 5 mph) as the speed limit? Are there any road signs that require your reaction? If any, what are they?
L3 Projection	If no action is taken, are there any pedestrians that may collide with you? If any, please select their locations from the given picture.	Is the gap from the car ahead of you getting smaller, larger, or remaining the same? If no action is taken, are there any vehicles that may collide with you? If any, please select their locations from the given picture	Do you need to make a turn at the upcoming intersection to get to the destination?	Can you make it through the intersection without stopping?

Note. SA = situation awareness.

TABLE A2: Effects of Visual Warnings on Driver Gaze Dwell Time, SA, and OC

Dependent Variables	Pedestrian	Moving Obstacle	Stationary Obstacles	Legal Obstacles
Gaze Dwell Time	$F(2, 42) = 101.39$, $p = .000^*$	$F(2, 42) = 1.86$, $p = .168$	$F(2, 42) = 2.53$, $p = .107$	$F(2, 42) = 4.10$, $p = .024^*$
SA, Perception	$Q(2, 23) = 10.71$, $p = .005^*$	$Q(2, 23) = 0.00$, $p = 1.000$	$Q(2, 23) = 0.00$, $p = 1.000$	$Q(2, 23) = 3.00$, $p = .223$
SA, Comprehension	$Q(2, 23) = 12.67$, $p = .002^*$	$Q(2, 23) = 0.00$, $p = 1.000$	$Q(2, 23) = 1.73$, $p = .420$	$Q(2, 23) = 8.14$, $p = .017^*$
SA, Projection	$Q(2, 23) = 10.11$, $p = .006^*$	$Q(2, 23) = 5.32$, $p = .070^{\#}$	$Q(2, 23) = 0.14$, $p = .931$	$Q(2, 23) = 6.50$, $p = .039^*$
OC, Perception	$Q(2, 23) = 10.80$, $p = .005^*$	$Q(2, 23) = 1.00$, $p = .607$	$Q(2, 23) = 0.00$, $p = 1.000$	$Q(2, 23) = 6.00$, $p = .050$
OC, Comprehension	$Q(2, 23) = 14.63$, $p = .001^*$	$Q(2, 23) = 1.00$, $p = .607$	$Q(2, 23) = 1.14$, $p = .565$	$Q(2, 23) = 5.25$, $p = .072^{\#}$
OC, Projection	$Q(2, 23) = 8.94$, $p = .011^*$	$Q(2, 23) = 6.20$, $p = .045^*$	$Q(2, 23) = 0.18$, $p = .913$	$Q(2, 23) = 4.00$, $p = .135$

Note. Results of analysis of variance (ANOVA) and Cochran's Q tests are reported with significance of the effects.

SA = situation awareness; OC = overconfidence.

* $p < .05$. $^{\#}p < .1$.

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KEY POINTS

- Augmented reality (AR) head-up displays (HUDs) can direct the driver's attention away from critical road elements that are not augmented by the displays.
- Current in-vehicle device assessment methods cannot capture the distraction potential of AR HUDs that do not take the driver's eyes off the road and mind off the driving.
- The visual distraction potential of AR HUDs was quantified by investigating driver gaze distribution across different types of objects on the road. The cognitive distraction was quantified by measuring driver awareness of objects on the road that are not augmented by the display.
- A human-subject experiment in a driving simulator demonstrated the potential of the proposed method and showed that augmented reality driver

interfaces can be either informative or distractive depending upon how we design perceptual forms of graphical elements presented on the HUD.

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